



DEPARTMENT OF MATHEMATICS AND COMPUTING
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. Second Semester CE/EE/SET/ECE/ME/R&IA)

Mid Term – II Examination

Integral Calculus and Difference Equations (MAIC 102)

Max. Marks: 15 Time: 10:00 am – 10:50 am

Note: Answer all the questions.
Calculator is not allowed into the examination hall.

- Q1. Evaluate $\iint_R (x + y)^2 dx dy$, where R is the region parallelogram in the xy – plane with vertices $(1, 0)$, $(3, 1)$, $(2, 2)$, $(0, 1)$, using the transformation $u = x + y$ and $v = x - 2y$. (5M)
- Q2. Find the directional derivative of $f(x, y, z) = x^2 y^2 z^2$ at the point $(1, 1, -1)$ in the direction of the tangent to the curve $x = e^t$, $y = \sin 2t + 1$, $z = 1 - \cos t$ at $t = 0$. (5M)
- Q3. State and verify Gauss divergence theorem for $\vec{F} = 4x \hat{i} + 2y^2 \hat{j} + \hat{k}$ taken over the region bounded by the cylinder $x^2 + y^2 = 4$, $z = 0$, $z = 3$. (5M)